# Suraj Kumar Choudhary

HPC-AI Engineer | System Administrator | AI/ML Developer

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## PROFESSIONAL SUMMARY

Results-driven HPC-AI Engineer with hands-on experience managing India's fastest supercomputers (Airawat and PARAM Siddhi AI) at C-DAC ACTS Pune, contributing to systems with 17.5 PF peak compute capacity. Skilled in HPC cluster administration, SLURM workload management, Linux systems, AI/ML model development, and full-stack web development. Passionate about applying machine learning to predictive maintenance, anomaly detection, and high-performance computing optimization.

## WORK EXPERIENCE

**HPC-AI Engineer (Project Engineer)** | **C-DAC ACTS Pune** *Mar 2025 – Present*

*National Supercomputing Mission (NSM) | NPSF Team*

- Manage and maintain India's fastest supercomputers — Airawat and PARAM Siddhi AI — ensuring stable operation at 17.5 PF peak compute capacity.
- Administer SLURM workload manager for efficient job scheduling, resource allocation, and queue management across large-scale HPC clusters.
- Monitor cluster health using Prometheus and Grafana dashboards; proactively detect and resolve node/network failures to maintain high availability.
- Contributing to the SDDG PoC (Secure Distributed Data Grid) project under NSM, enabling high-speed 400G data transfer between HPC sites (Pune–Bangalore) for secure job migration and resource utilization optimization.
- Perform routine system administration tasks: OS patching, firmware updates, storage management, and user account provisioning on Linux-based HPC nodes.

**AI/ML Intern** | **NIELIT (Work-Based Learning)** *Feb 2024 – Jul 2024*

- Developed and trained machine learning models for classification and prediction tasks using Python, Scikit-learn, and TensorFlow.
- Conducted data preprocessing, feature engineering, and model evaluation to improve accuracy and generalization on real-world datasets.
- Gained practical exposure to deep learning architectures, computer vision techniques, and model deployment workflows.

**Front-End Developer Intern** | **LadyBird Web Solution Pvt. Ltd.** *Jun 2023 – Aug 2023*

- Designed and developed responsive web interfaces using React.js, JavaScript, HTML5, and CSS3.
- Collaborated with backend teams to integrate RESTful APIs and deliver dynamic user-facing features within Agile sprint cycles.
- Improved UI performance and cross-browser compatibility; contributed to component library development.

## PROJECTS

**SDDG POC — Secure Distributed Data Grid (NSM)** · *HPC · Networking · Linux · 400G Interconnect*

- Leading a Proof-of-Concept for high-speed secure data transfer between HPC clusters (Pune–Bangalore) under the National Supercomputing Mission.
- Focused on enabling efficient job migration, secure multi-site data movement, and optimized resource utilization across geographically distributed clusters at 400 Gbps throughput.

### HPC Node/Cluster Failure Prediction with Prevention · *Python · Prometheus · Grafana · ML · Shell · SLURM* [GitHub]

- Designed and implemented an AI-powered predictive maintenance system analyzing real-time HPC metrics (CPU, memory, latency, disk I/O) from Prometheus and Grafana.
- Trained ML models for anomaly detection and failure prediction, significantly reducing unplanned downtime and improving cluster reliability.
- Automated alerting and prevention workflows via Shell scripts integrated with SLURM job scheduler.

### Vehicle Detection & Counting System · *Python · YOLOv5 · Deep SORT · OpenCV · Streamlit* [GitHub]

- Built a real-time computer vision application to detect, classify, track, and count vehicles from video feeds using YOLOv5 for object detection and Deep SORT for multi-object tracking.
- Developed an interactive Streamlit dashboard for visualization and reporting of vehicle analytics.

### Stock Price Predictor Web App · *Python · Machine Learning · Streamlit · Jupyter* [GitHub]

- Built an ML-powered stock price prediction web application integrating real-time market data with predictive models to deliver actionable trading insights.

### IPL Match Win Probability Predictor · *Python · ML · Streamlit · Jupyter* [GitHub]

- Developed a Streamlit web app using machine learning to predict IPL match win probabilities based on live inputs: teams, current score, overs, and wickets.

### RecipeFusion Web App · *React.js · JavaScript · React Router* [GitHub]

- Built a React-based recipe discovery app with seamless navigation via React Router, category-based browsing, and detailed recipe views.

## TECHNICAL SKILLS

**HPC & Systems:** SLURM, PBS/Torque, HPC Cluster Administration, Linux (RHEL/CentOS/Ubuntu), Bash Scripting, System Administration

**AI / ML:** Python, Scikit-learn, TensorFlow, Keras, YOLOv5, Deep SORT, OpenCV, Anomaly Detection, Predictive Modeling

**Cloud & DevOps:** Docker, Kubernetes, AWS (EC2, S3), Prometheus, Grafana, CI/CD

**Web Development:** React.js, JavaScript (ES6+), HTML5, CSS3, Streamlit, REST APIs

**Networking & Storage:** High-Speed Interconnects (400G), InfiniBand, Parallel File Systems, Data Grid Computing

**Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code, Zabbix

## EDUCATION

**PG Diploma in HPC System Administration (PG-DHPCSA)** *Aug 2024 – Mar 2025*
C-DAC ACTS Pune, Maharashtra

**Bachelor of Engineering (B.E.)** *Jul 2020 – Aug 2024*
Mahant Bachittar Singh College of Engineering and Technology, Jammu

## CERTIFICATIONS & TRAINING

- PG-DHPCSA — C-DAC ACTS Pune (HPC System Administration, SLURM, Linux, Parallel Computing)
- Work-Based Learning (WBL) in AI/ML — NIELIT, Jammu
- Front-End Web Development — LadyBird Web Solution Pvt. Ltd.

## LANGUAGES

**English** — Professional Working Proficiency | **Hindi** — Native / Bilingual Proficiency